NEP Aligned

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Data Analytics with R

Quality Laboratory Manual

<u>Experiment No. 10</u>

To connect Power BI to an external dataset, clean and transform data and create interactive dashboards using filters, slicers and visual charts

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Experiment No. 09

Title of Experiment: To connect Power BI to an external dataset, clean and transform data and create interactive dashboards using filters, slicers and visual charts

Aim of Experiment: To connect Power BI with an external dataset, perform data cleaning and transformation operations and design an interactive dashboard using filters, slicers and visualization charts for analytical reporting

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Built-in R graphics package
- Internet Connection (for dataset access)/ CSV dataset file

Experiment Objectives:

- To connect Power BI to external datasets and import CSV/Excel data into Power BI
- To perform data cleaning and preprocessing
- To apply transformation and create visualizations using Power Query Editor
- To use slicers and filter for interactive analysis and design analytical dashboards for decision-making

Experiment Outcomes:

After completing the experiment, students will be able to –

- Connect and import external datasets into Power BI
- Clean and transform raw data effectively
- Build interactive dashboards and use visual analytics for business insights
- Create professional reports for decision-making

Theory:

Power BI is a business intelligence and data visualization tool developed by Microsoft. It allows users to connect to various data sources, transform raw data, create interactive visualizations, and generate dashboards for business analytics.

Power BI is widely used in:

- Business intelligence
- Sales analytics
- Financial reporting
- Healthcare analytics
- Educational analytics
- Manufacturing analytics

It enables organizations to convert raw data into meaningful insights.

Connecting to External Datasets

Power BI supports multiple data sources such as:

- Excel files
- CSV files
- SQL databases
- Cloud services
- Web APIs

The "Get Data" feature is used to import external datasets into Power BI Desktop.
Steps:

- Open Power BI Desktop
- Click Get Data
- Select source type
- Load dataset

Data Cleaning and Transformation

Raw datasets often contain:

- Missing values
- Duplicate records
- Incorrect column names
- Data type mismatches

Power BI provides Power Query Editor for preprocessing tasks such as:

- Removing duplicates
- Renaming columns
- Changing data types
- Filtering rows
- Splitting columns
- Handling null values

Data transformation improves data quality and analytical accuracy

Interactive Dashboards

A dashboard is a visual representation of important information and KPIs. Power BI dashboards combine multiple visual elements into a single interactive interface.

Common dashboard visuals:

- Bar charts
- Pie charts
- Line charts
- Tables
- Cards
- Maps

Interactive dashboards help users explore insights dynamically

Filters and Slicers

Filters

Filters limit data displayed in visuals.

Types:

- Visual-level filters
- Page-level filters
- Report-level filters

Slicers

Slicers provide interactive filtering options for users.

Examples:

- Region-wise filtering
- Year-wise filtering
- Product category filtering

Slicers improve dashboard interactivity and user experience

Visualization Charts

Power BI supports various visualization charts:

Chart Type	Purpose
Bar Chart	Category comparison
Pie Chart	Percentage distribution
Line Chart	Trend analysis
Area Chart	Time-series trends
Scatter Plot	Relationship analysis
KPI cards	Performance indicators

Selecting appropriate visualizations improves analytical interpretation

Importance of Dashboards in Analytics

Dashboards provide:

- Real-time monitoring
- Data-driven decision-making

- Trend analysis
- Performance tracking
- Interactive exploration

Organizations use dashboards to simplify complex business data

Task to Perform

The reference sheet has been shared on the Moodle LMS along with this experiment. Each student has been assigned a unique topic to create a Power BI dashboard. All students should prepare the dashboard accordingly and submit it

Conclusion:
The experiment successfully demonstrated how to connect Power BI to an external dataset, clean and transform data, and create interactive dashboards using filters, slicers, and visual charts. Power BI enables effective business intelligence reporting and helps organizations make data-driven decisions through interactive analytics

Expected Oral Questions:
1) What is Power BI?
2) What is Power Query Editor?
3) Difference between filter and slicer?
4) Why is data cleaning important?
5) What is a dashboard?
6) Which charts are suitable for trend analysis?
7) What are KPI cards?
8) How do slicers improve dashboards?
9) What is ETL process?
10) Why are dashboards important in business analytics?

FAQs in Interview:
1) Why is Power BI popular in industry?
2) What is the role of Power Query?
3) What is the difference between dashboard and report?
4) Why are slicers important?
5) What types of data sources can Power BI connect to?